Tutorial - 7 (KNN) 17/3/26

Classification

Training Dataset :-

Action scenes	Romance scenes	Happy Ending	Genre
15	2	0	A
1	12	1	R
18	1	0	A
2	14	1	R
14	3	0	A

Test Data

Test ID	Action scenes	Romance Scenes	Happy Ending	Actual Genre
T_1	10	5	1	A

(10, 5, 1) — (15, 2, 0)

(1, 12, 1)

(18, 1, 0)

(2, 14, 1)

(14, 3, 1)

$$\text{distance} = \sqrt{(x_{test1} - x_1)^2 + (x_{test2} - x_2)^2 + (x_{test3} - x_3)^2}$$

$x_1 \; y_1 \; z_1$
$(10, 5, 1)$

$(15, 2, 0)$
$x_2 \; y_2 \; z_2$

$$= \sqrt{(10-15)^2 + (5-2)^2 + (1-0)^2}$$

$$= \sqrt{25 + 9 + 1} = \sqrt{35} = 5.92$$

$$(10,5,1)(1,12,1) = \sqrt{(10-1)^2 + (5-12)^2 + (1-1)^2}$$

$$= \sqrt{(9)^2 + (-7)^2} = \sqrt{81+49} = \sqrt{130} = 11.40$$

$$(10,5,1)(18,1,0) = \sqrt{(10-18)^2 + (5-1)^2 + (1-0)^2}$$

$$= \sqrt{(-8)^2 + (4)^2 + (1)^2}$$

$$= \sqrt{64+16+1} = \sqrt{81} = 9.0$$

$$(10,5,1)(2,14,1) = \sqrt{(10-2)^2 + (5-14)^2 + (1-1)^2} = \sqrt{8^2 + 9^2}$$

$$= \sqrt{64+81} = \sqrt{145} = 12.04$$

$$(10,5,1)(14,3,1) = \sqrt{(10-14)^2 + (5-3)^2 + (1-1)^2}$$

$$= \sqrt{4^2 + 2^2 + 0} = \sqrt{16+4} = \sqrt{20} = 4.47$$

they will give

$\hookrightarrow K = 3$

$d_1 = 5.92 \rightarrow A$

$d_3 = 9.0 \rightarrow A$

$d_5 = 4.47 = A$

Actual and predicated value Grade is A
so predicated value

$k=3$ $d_1 = 5.92$ weight $= \frac{1}{d} = \frac{1}{5.92} = 0.19$

$d_3 = 9.0$ weight $= \frac{1}{d} = \frac{1}{9.0} = 0.11$

$d_5 = 4.47$ weight $= \frac{1}{d} = \frac{1}{4.47} = 0.22$

weight of A $= w_1 + w_3 + w_5$
$= 0.52$

$k=5$ A $(w_1 = 0.19)$
$d_2 = 11.40 \rightarrow$ R $(w_2 = 0.08)$
$d_3 = 9.0 \rightarrow$ A $(w_3 = 0.11)$
$d_4 = 12.04 \rightarrow$ R $(w_4 = 0.8)$
$d_5 = 4.47 \rightarrow$ A $(w_5 = 0.22$

weight of A $= w_1 + w_2 + w_3 = 0.52$

weight of B $= w_2 + w_4 = 0.16$

Regression

Age (x_1)	previous value (x_2) in lakh	car price (y) in lakh
2	3	2
3	4	3
5	1	0.5

Test : $(4, 2, y)$

$(4,2) \leftarrow (2,3)$

$(4,2) \leftarrow (3,4)$

$(4,2) \leftarrow (5,1)$

$\overset{x_1}{(4}, \overset{y_1}{2)} \overset{x_2}{(2}, \overset{y_2}{3)} = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2} = \sqrt{(4-2)^2+(2-3)^2} = \sqrt{5}$
$= 2.24$

$(4,2)(3,4) = \sqrt{(4-3)^2+(2-4)^2} = \sqrt{5} = 2.24$

$(4,2)(5,1) = \sqrt{(4-5)^2+(2-1)^2} = \sqrt{2} = 1.41$

Give that in question $\boxed{K=2}$

$d_1 = 2.24$

$d_2 = 2.24 \rightarrow 3$

$d_3 = 1.41 \rightarrow 0.5$

$$\boxed{\hat{y} = \frac{1}{K}\sum_{i=1}^{K} y_i}$$

$$= \frac{3+0.5}{2} = 1.75$$

$= 1.75$ Test $= (4, 4, \overset{y}{5}) \rightarrow 1.75$